

BreezeBnb

BreezeBnb connects travelers with cozy, unique stays in local homes, offering a comfortable and memorable experience. Whether it's a city getaway or a rural retreat, BreezeBnb aims to bring the world closer, one stay at a time. After five successful years managing data in excel, they've decided to create a data platform that delivers business intelligence to their market teams. This new platform aims to enhance decision-making and streamline insights. They provide stays only in New York city and intend to **expand to 20 new cities over the next year.**

Objective:

Build a simple data engineering pipeline to provide the following insights to the business users, considering the company's expansion plans.

Expectations:

1. A summary that helps business users understand key insights like:
 - a. Average price per night by neighborhood each month
 - b. Top 10 over/under priced listings as compared to the average for that neighbourhood.
- NOTE: These are just 2 examples of insight your solution should enable.**
2. Share a (< 500 words) view on what Production deployment looks like.

Please call out all assumptions beyond what has been stated in the problem statement

Current system capabilities:

1. An orchestrator which supports running user defined jobs via [crontab](#) file specified within the code repository. It effectively schedules the job automatically based on the schedule defined
2. End-users will provide the required datasets under the dataset directory:
 - a. [listings.csv.gz](#) containing fields as described in the [data dictionary](#).
 - i. Provided once each month

Please make assumptions as necessary and call these out clearly as part of your solution.

Acknowledgment

The linked listings dataset is from [Inside Airbnb](#)

General Guidelines:

At Sahaj, we strive to build high-quality software that has strong aesthetics (is readable and maintainable), has extensive safety nets to safeguard quality, handles errors gracefully and works as expected, without breaking down, with varying input.

We are looking for people who can write code that has flexibility built-in, by adhering to the principles of clean coding and good data engineering practices, and have the ability to deal with the real-life constraints/trade-offs while designing a system.

*It is important to note that we are not looking for a full-fledged data engineering pipeline inclusive of infrastructural concerns, and a massive inventory of tools. **The focus is on the overall design.** So, while building a solution, it would be nicer if the input to the code is provided either via unit tests or a file. Using the command line (for input/output) can be tedious and difficult to test, so it is best avoided.*

Following is a list of things to keep in mind, before you submit your code, to ensure that your code focuses on attributes, we are looking for -

- Have you applied the principles of [YAGNI](#) and [KISS](#) (additional info [here](#))?
- Have you unit tested your code or did TDD? If you have not, we strongly recommend you read about it and attempt it with your solution. We like to be confident in what we deliver and want to see similar evidence in the solution. This is usually in the form of tests, however we are open to learn your perspectives on the same. Submissions without guarantees that they work as expected are generally frowned upon.
- Have you looked at basic refactoring to improve the design of your code? [Here](#) are some guidelines for the same.
- Finally, and foremost, are the principles applied in a pragmatic way. Simplicity is the strongest of the traits of a piece of code. However, easily written code may not necessarily be simple code.
- Use of any frameworks, infrastructure aspects is unnecessary and will complicate the solution. It will also extend the time taken to solve the problem. Prefer focussing on the core problem statement and use libraries where necessary

Please send the code back to us in a zip file. Please do not add the data files in it to keep the zip file size small. Alternatively, If you would like to share a git repository, please ensure that [it is private](#) and the user "sahaj-org-building" needs to be [invited as a collaborator on your project](#).